

# LAB EXPERIMENT RECORD

## Set 11: Product Category Analysis

|                           |                                           |
|---------------------------|-------------------------------------------|
| Department                | AIML                                      |
| Course Code / Course Name | DSA0601 / Data Handling and Visualization |
| Experiment                | Data Visualization using R and Tableau    |
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| Date                      | 05 October 2026                           |
| Day                       | Monday                                    |

### Aim

To analyze product category sales using a funnel chart, tabular representation, Tableau dashboard, and pie chart, and to visualize the distribution of sales across product categories.

### Dataset: Product Categories

| Category    | Sales (in \$) |
|-------------|---------------|
| Electronics | 50000         |
| Clothing    | 35000         |
| Appliances  | 40000         |

### PROGRAM 1: Funnel Chart for Product Category Sales

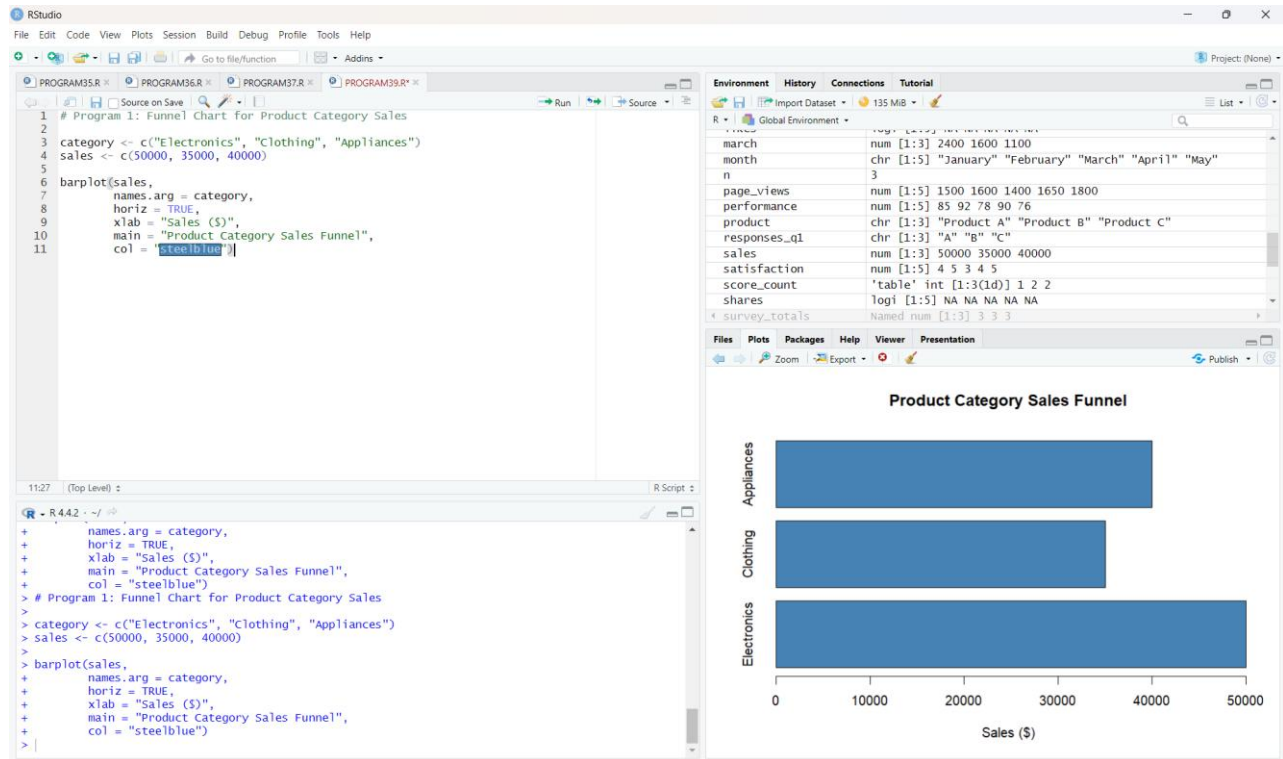
Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

```
# Program 1: Funnel Chart for Product Category Sales
```

```
category <- c("Electronics", "Clothing", "Appliances")
sales <- c(50000, 35000, 40000)
```

```
barplot(sales,
        names.arg = category,
        horiz = TRUE,
        xlab = "Sales ($)",
        main = "Product Category Sales Funnel",
        col = "steelblue")
```

## Output 1: Product Category Sales Funnel



Output 1: Funnel Chart for Product Category Sales

Observation: The funnel-style horizontal bar chart compares sales across the three product categories. Electronics has the highest sales at \$50,000, followed by Appliances at \$40,000 and Clothing at \$35,000.

## PROGRAM 2: Product Category Sales Table

Build a table to display the sales data for each product category. Label the table elements.

```
# Program 2: Product Category Sales Table
```

```
product_data <- data.frame(  
  Category = c("Electronics", "Clothing", "Appliances"),  
  Sales = c(50000, 35000, 40000)  
)  
  
print(product_data)
```

## Output 2: Product Category Sales Table

The screenshot shows the RStudio interface. The script editor on the left contains the following code:

```
1 # Program 2: Product Category Sales Table
2
3 product_data <- data.frame(
4   Category = c("Electronics", "Clothing", "Appliances"),
5   Sales = c(50000, 35000, 40000)
6 )
7
8 print(product_data)
```

The Environment pane on the right shows the objects created:

- `product_data`: 3 obs. of 2 variables
- `sales_data`: 3 obs. of 4 variables
- `survey_data`: 3 obs. of 4 variables
- `ui`: List of 4

The console at the bottom shows the output of the `print` function:

```
> # Program 2: Product Category Sales Table
>
> product_data <- data.frame(
+   Category = c("Electronics", "Clothing", "Appliances"),
+   Sales = c(50000, 35000, 40000)
+ )
>
> print(product_data)
  Category Sales
1 Electronics 50000
2  Clothing  35000
3 Appliances  40000
```

Output 2: Product Category Sales Table

Observation: The table clearly displays each product category and its corresponding sales value.

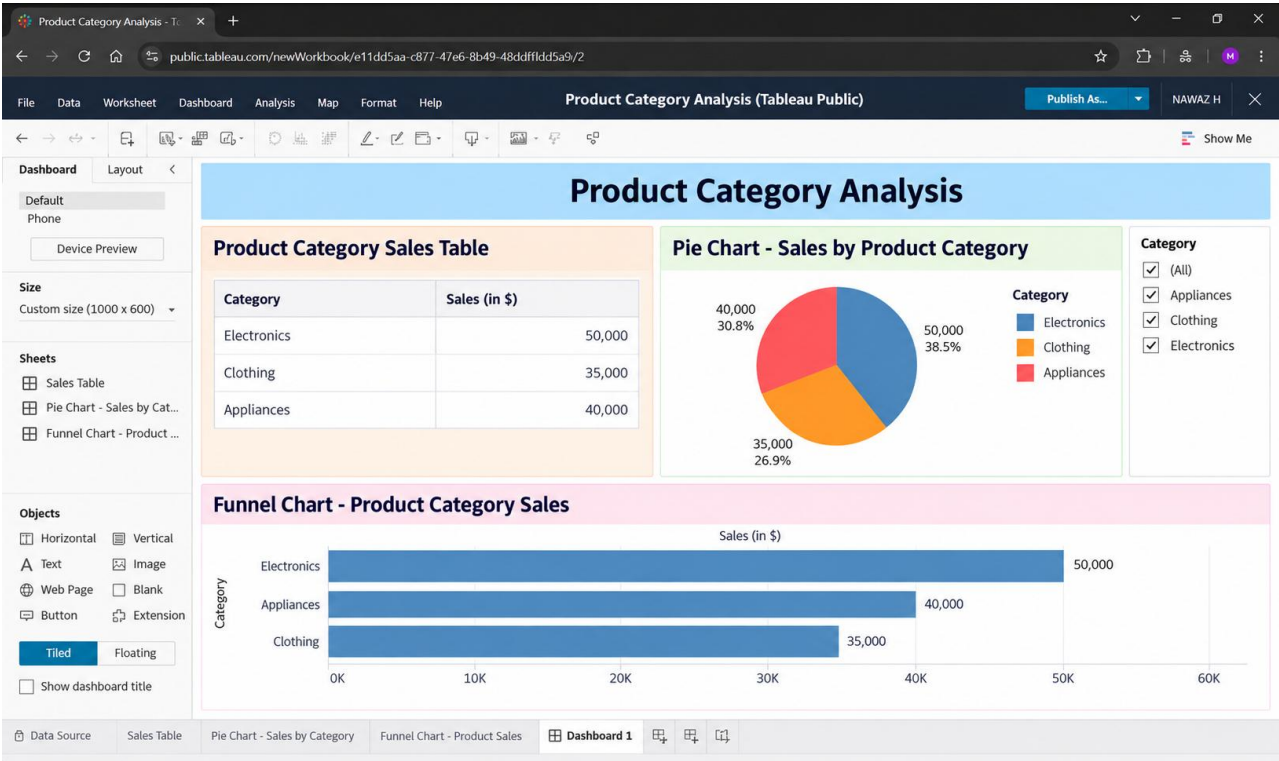
## PROGRAM 3: Tableau Dashboard

Develop a Tableau dashboard combining the pie chart, funnel chart, and table for interactive exploration of product category data.

### TABLEAU DASHBOARD STEPS

1. Enter the Product Categories dataset into Tableau Public.
2. Create a Sales Table worksheet using Category and Sales.
3. Create a Pie Chart using Category and Sales.
4. Create a Funnel Chart using Category and Sales as a horizontal bar chart.
5. Create a new dashboard and place the Sales Table, Pie Chart, and Funnel Chart together.
6. Add the Category filter for interactive exploration.
7. Set the dashboard title to "Product Category Analysis".

### Output 3: Tableau Product Category Analysis Dashboard



Output 3: Tableau Product Category Analysis Dashboard

Observation: The Tableau dashboard combines the sales table, pie chart, and funnel chart in one interactive view. The Category filter allows individual product categories to be explored.

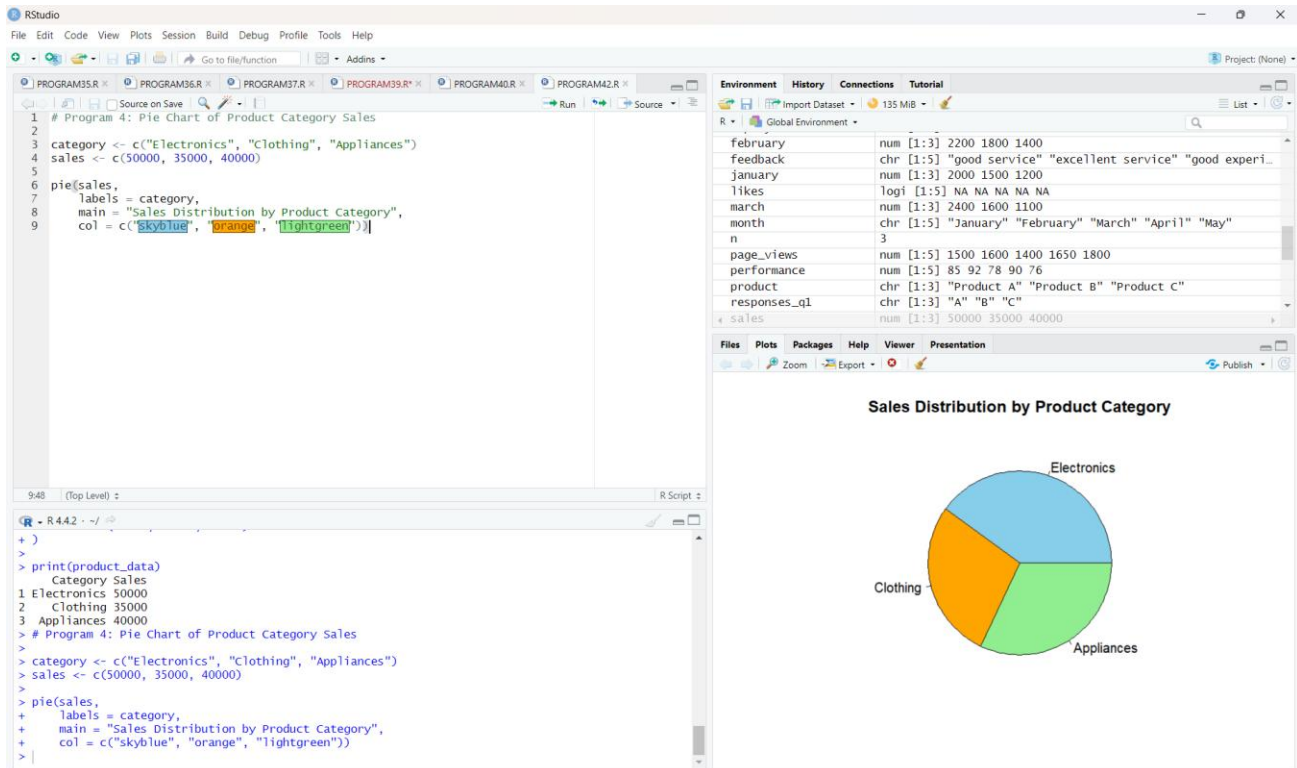
## PROGRAM 4: Pie Chart of Product Category Sales

Create a pie chart to represent the distribution of sales across product categories. Include labels.

# Program 4: Pie Chart of Product Category Sales

```
category <- c("Electronics", "Clothing", "Appliances")
sales <- c(50000, 35000, 40000)
```

```
pie(sales,
    labels = category,
    main = "Sales Distribution by Product Category",
    col = c("skyblue", "orange", "lightgreen"))
```



Output 4: Sales Distribution by Product Category

Observation: The pie chart shows the share of total sales contributed by each category. Electronics contributes 40.0%, Appliances 32.0%, and Clothing 28.0%.

### Procedure

1. Create the product category and sales vectors in R.
2. Use `barplot()` with horizontal orientation to create the funnel-style product category sales chart.
3. Create and print the product category sales table using `data.frame()`.
4. Enter the product category sales data into Tableau and create the Sales Table, Pie Chart, and Funnel Chart worksheets.
5. Combine the Tableau worksheets into a dashboard, add the Category filter, and format the dashboard for interactive exploration.
6. Create the pie chart in R using `pie()` and include category labels.

### Result

The Product Category Analysis was successfully completed using R and Tableau. The sales data was represented through a funnel chart, table, interactive Tableau dashboard, and pie chart.

### Key R Functions Used

| Function                  | Purpose                                          |
|---------------------------|--------------------------------------------------|
| <code>barplot()</code>    | Creates the horizontal funnel-style sales chart. |
| <code>data.frame()</code> | Creates the product category sales table.        |
| <code>pie()</code>        | Creates the pie chart for sales distribution.    |
| <code>print()</code>      | Displays the product category sales data.        |